

Matrix Theory Lecture Notes

Compiled for Exam Preparation

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1 Basics of Matrix and Determinant

Series: Matrix Theory Lecture: 01 YouTube: <https://www.youtube.com/watch?v=2NRTmf4SrII>

1.1 Prerequisites

- Basic algebra and arithmetic operations.
- Familiarity with basic mathematical notations.

1.2 Core Content

A matrix is a rectangular array of numbers arranged in rows and columns. Matrices provide a compact way to represent and work with linear transformations and systems of equations.

1.2.1 Definition of a Matrix

An $m \times n$ matrix contains m rows and n columns. The elements are denoted by a_{ij} , representing the entry in the i -th row and j -th column.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

1.2.2 Types of Matrices

- **Square Matrix:** A matrix with an equal number of rows and columns ($m = n$).
- **Rectangular Matrix:** A matrix where the number of rows is not equal to the number of columns ($m \neq n$).
- **Diagonal Matrix:** A square matrix where all off-diagonal elements are zero.
- **Identity Matrix:** A diagonal matrix where all main diagonal elements are equal to 1.

1.2.3 Matrix Determinant

The determinant is a scalar value calculated from a square matrix. It provides important properties about the matrix, such as invertibility. A matrix with a non-zero determinant is non-singular and invertible.

For a 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The determinant is:

$$|A| = ad - bc$$

For a 3×3 matrix, expansion by minors along the first row gives:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

The determinant calculation follows:

$$|A| = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

1.2.4 Matrix Operations

- **Addition:** Add corresponding elements of matrices of the same dimensions.
- **Scalar Multiplication:** Multiply every element of the matrix by a constant scalar.
- **Matrix Multiplication:** The sum of products of elements from rows of the first matrix and columns of the second matrix. This requires the number of columns in the first matrix to match the number of rows in the second.

1.3 What Comes Next

2 Rank of Matrix A Concept

Series: Matrix Theory **Lecture:** 02 **YouTube:** <https://www.youtube.com/watch?v=h1FgjMdIEJ0>

2.1 Prerequisites

- Understanding of square matrices and determinants.
- Familiarity with expansion of determinants by minors.

2.2 Core Content

The rank of a matrix is a fundamental property that provides insight into the linear independence of its rows and columns. It is central to solving systems of linear equations.

2.2.1 Definition of a Minor

Consider an $m \times n$ matrix A . If we retain any r rows and r columns from A and delete the rest, the determinant of the resulting $r \times r$ submatrix is called a minor of A of order r .

2.2.2 Definition of Rank

The rank of a matrix A is denoted by $\rho(A)$. A number r is the rank of matrix A if it satisfies two conditions: 1. There exists at least one non-zero minor of order r . 2. Every minor of order greater than r is equal to zero. If A is a non-zero matrix, then its rank is at least 1. The rank of a zero matrix is defined as 0.

2.2.3 Finding Rank using Minors

To find the rank of a matrix using the minor method, follow these steps: 1. Start with the highest possible order of a square submatrix. 2. Calculate the determinant (minor) of this submatrix. 3. If the determinant is non-zero, the rank is the order of this submatrix. 4. If all minors of this highest order are zero, step down to the next lower order and check its minors. 5. The rank is the order of the highest order non-zero minor.

2.2.4 Nullity

The nullity of a square matrix of order n is defined as the difference between its order and its rank.

$$N(A) = n - \rho(A)$$

Nullity represents the dimension of the null space of the matrix.

2.3 What Comes Next

3 Rank of Matrix By Determinant and Echelon Form

Series: Matrix Theory **Lecture:** 03 **YouTube:** <https://www.youtube.com/watch?v=DNFrFWb7RpM>

3.1 Prerequisites

- Understanding of matrix rank and minors.
- Familiarity with elementary row operations.

3.2 Core Content

Calculating the rank of a large matrix by evaluating all possible minors becomes computationally expensive. We use systematic methods like Echelon form to simplify this process.

3.2.1 Elementary Row Operations

We can transform a matrix into a simpler, equivalent form without changing its rank. The valid elementary row operations are: 1. Interchanging any two rows ($R_i \leftrightarrow R_j$). 2. Multiplying a row by a non-zero scalar ($R_i \rightarrow kR_i$). 3. Adding a scalar multiple of one row to another row ($R_i \rightarrow R_i + kR_j$).

Two matrices are called equivalent ($A \sim B$) if one can be obtained from the other through a sequence of elementary row operations. Equivalent matrices have the same rank.

3.2.2 Echelon Form

A matrix is in Echelon form if it satisfies the following conditions: 1. All non-zero rows are above any rows of all zeroes. 2. The leading coefficient (the first non-zero number from the left, also called the pivot) of a non-zero row is always strictly to the right of the leading coefficient of the row above it. 3. All entries in a column below a leading coefficient are zeroes.

3.2.3 Rank using Echelon Form

To find the rank of a matrix using Echelon form: 1. Apply elementary row operations to transform the matrix into Echelon form. 2. Count the number of non-zero rows in the resulting matrix. 3. The rank of the matrix is equal to the number of non-zero rows.

This method is highly systematic and preferred over the minor method.

3.3 Solved Example

Let us find the rank of matrix A :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$$

Apply row operations to make entries below the first pivot zero:

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

This gives the equivalent matrix:

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & -2 & -4 \end{bmatrix}$$

Apply row operation to make entry below the second pivot zero:

$$R_3 \rightarrow R_3 - 2R_2$$

This gives the matrix in Echelon form:

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

There are two non-zero rows in the Echelon form matrix. Therefore, the rank is $\rho(A) = 2$.

3.4 What Comes Next

4 Rank of Matrix By Normal / Canonical Form

Series: Matrix Theory **Lecture:** 04 **YouTube:** <https://www.youtube.com/watch?v=gXwEtwX8g8M>

4.1 Prerequisites

- Knowledge of matrix rank.
- Understanding of elementary row and column operations.

4.2 Core Content

The Normal form, also known as the Canonical form, is a standard form used to determine the rank of a matrix. Unlike the Echelon form which only uses row operations, reducing a matrix to Normal form involves applying both elementary row and elementary column operations.

4.2.1 Normal Form Definition

Every non-zero matrix A of rank r can be reduced, by a sequence of elementary row and column operations, to one of the following forms:

$$\begin{bmatrix} I_r & O \\ O & O \end{bmatrix}, \quad \begin{bmatrix} I_r \\ O \end{bmatrix}, \quad [I_r \ O], \quad \text{or} \quad I_r$$

Here, I_r represents an identity matrix of order r , and O represents a zero matrix. The value r gives the rank of the original matrix A .

4.2.2 Procedure for Normal Form Reduction

1. Locate a non-zero element in the matrix. If the element a_{11} is not 1, interchange rows or columns to bring a 1 to the $(1,1)$ position. This simplifies calculations.
2. Use elementary row operations to make all other elements in the first column zero.
3. Use elementary column operations to make all other elements in the first row zero.
4. Move to the next diagonal element a_{22} . Make it 1 and use row and column operations to clear out the rest of the second column and second row.
5. Repeat this process down the main diagonal until the matrix is reduced to the identity block I_r and zeros everywhere else.

4.2.3 Advantages of Normal Form

- It provides a definitive, clear-cut identification of rank. The rank is simply the order of the identity submatrix generated.
- It is useful for finding the equivalence matrices P and Q such that PAQ is in Normal form.

4.3 Solved Example

Let us reduce matrix A to Normal form to find its rank:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 5 & 0 \\ 3 & 7 & -1 \end{bmatrix}$$

Use row operations to clear the first column below the $(1,1)$ pivot:

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$A \sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

Use column operations to clear the first row to the right of the $(1, 1)$ pivot:

$$C_2 \rightarrow C_2 - 2C_1$$

$$C_3 \rightarrow C_3 + C_1$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$

Use row operation to clear below the $(2, 2)$ pivot:

$$R_3 \rightarrow R_3 - R_2$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Use column operation to clear to the right of the $(2, 2)$ pivot:

$$C_3 \rightarrow C_3 - 2C_2$$

$$A \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

This is the Normal form block:

$$\begin{bmatrix} I_2 & O \\ O & O \end{bmatrix}$$

The identity block is of order 2. Therefore, the rank is $\rho(A) = 2$.

4.4 What Comes Next

5 Consistency of System of Linear Equations

Series: Matrix Theory **Lecture:** 05 **YouTube:** <https://www.youtube.com/watch?v=bAJLtUzhBQI>

5.1 Prerequisites

- Proficiency in calculating the rank of a matrix.
- Understanding of Echelon form reductions.

5.2 Core Content

A system of linear equations is a collection of one or more linear equations involving the same set of variables. We use matrix theory to determine whether a system has a solution (consistent) or no solution (inconsistent).

5.2.1 System Representation

Consider a system of m non-homogeneous linear equations with n variables $x_1, x_2, \dots, x_n$:

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m\end{aligned}$$

This system can be written in matrix form as $AX = B$:

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

Where: - A is the coefficient matrix. - X is the column vector of variables. - B is the column vector of constants.

5.2.2 The Augmented Matrix

The augmented matrix $[A : B]$ is formed by appending the constant vector B as an additional column to the coefficient matrix A .

$$[A : B] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array} \right]$$

5.2.3 Conditions for Consistency (Rouché-Capelli Theorem)

To analyze consistency, we reduce the augmented matrix $[A : B]$ to Echelon form and determine the rank of the coefficient matrix $\rho(A)$ and the rank of the augmented matrix $\rho([A : B])$.

There are three possible outcomes: 1. **Inconsistent System (No Solution):** If $\rho(A) \neq \rho([A : B])$. The system has no valid solution. 2. **Consistent System (Unique Solution):** If $\rho(A) = \rho([A : B]) = n$, where n is the number of variables. The system yields exactly one unique solution for each variable. 3. **Consistent System (Infinite Solutions):** If $\rho(A) = \rho([A : B]) = r < n$. The system has infinitely many solutions. We must choose $n - r$ variables arbitrarily as free parameters (e.g., k_1, k_2) and solve for the remaining variables in terms of these parameters.

5.3 What Comes Next

6 Questions Consistency of System of Linear Equations

Series: Matrix Theory **Lecture:** 06 **YouTube:** <https://www.youtube.com/watch?v=sV3-43kkvSo>

6.1 Prerequisites

- Understanding of the conditions for consistency (Rouché-Capelli theorem).
- Proficiency in reducing augmented matrices to Echelon form.

6.2 Core Content

This lecture applies the consistency rules to solve practical systems of non-homogeneous linear equations.

6.2.1 Problem Solving Approach

1. Form the augmented matrix $[A : B]$ from the given system $AX = B$.
2. Apply elementary row operations to reduce $[A : B]$ to Echelon form.
3. Compare the rank of the coefficient matrix $\rho(A)$ with the rank of the augmented matrix $\rho([A : B])$.
4. Conclude the nature of the solution (unique, infinite, or none) based on the ranks and the number of variables n .
5. If consistent, perform back-substitution from the Echelon form equations to find the variable values.

6.2.2 Solved Problem: Unique Solution

Test the consistency and solve:

$$3x + y + 2z = 11$$

$$2x + 3y + z = 11$$

$$x + 2y + 3z = 14$$

The augmented matrix is:

$$[A : B] = \left[\begin{array}{ccc|c} 3 & 1 & 2 & 11 \\ 2 & 3 & 1 & 11 \\ 1 & 2 & 3 & 14 \end{array} \right]$$

After interchanging $R_1 \leftrightarrow R_3$ to bring 1 to the pivot and performing row reductions to Echelon form:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 14 \\ 0 & 1 & 5 & 17 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Here, $\rho(A) = 3$ and $\rho([A : B]) = 3$. The number of variables is $n = 3$. Since $\rho(A) = \rho([A : B]) = 3$, the system is consistent with a unique solution.

By back-substitution:

$$z = 3$$

$$y + 5(3) = 17 \implies y = 2$$

$$x + 2(2) + 3(3) = 14 \implies x = 1$$

The unique solution is $x = 1, y = 2, z = 3$.

6.2.3 Solved Problem: Infinite Solutions

Test the consistency and solve:

$$5x + 3y + 7z = 4$$

$$3x + 26y + 2z = 9$$

$$7x + 2y + 10z = 5$$

The augmented matrix is:

$$[A : B] = \left[\begin{array}{ccc|c} 5 & 3 & 7 & 4 \\ 3 & 26 & 2 & 9 \\ 7 & 2 & 10 & 5 \end{array} \right]$$

Reducing to Echelon form yields:

$$[A : B] \sim \left[\begin{array}{ccc|c} 1 & 49 & -3 & 14 \\ 0 & 11 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Here, $\rho(A) = 2$ and $\rho([A : B]) = 2$. Since $\rho(A) = \rho([A : B]) = 2 < 3$ (number of variables), the system has infinitely many solutions. We choose $3 - 2 = 1$ arbitrary variable. Let $z = k$.

$$11y - k = 3 \implies y = \frac{k + 3}{11}$$

Substituting back into the first row gives

$$x = \frac{7 - 16k}{11}.$$

6.3 What Comes Next

7 Concept of Eigen Values and Vectors

Series: Matrix Theory **Lecture:** 07 **YouTube:** <https://www.youtube.com/watch?v=GyT-SdFtqq4>

7.1 Prerequisites

- Familiarity with matrix determinants and polynomials.
- Ability to solve homogeneous systems of linear equations.

7.2 Core Content

Eigenvalues and eigenvectors are crucial concepts for analyzing linear transformations. They represent scalars and vectors that remain invariant in direction under a given matrix transformation.

7.2.1 Characteristic Matrix and Equation

Let A be a square matrix of order n , let λ be an unknown scalar variable, and let I be the identity matrix of the same order as A .

The matrix $A - \lambda I$ is called the **characteristic matrix** of A . The polynomial obtained by expanding the determinant

$$|A - \lambda I|$$

is called the **characteristic polynomial**. Equating the characteristic polynomial to zero gives the **characteristic equation**:

$$|A - \lambda I| = 0$$

7.2.2 Eigenvalues

The roots of the characteristic equation

$$|A - \lambda I| = 0$$

are called the eigenvalues (or characteristic roots, latent values) of matrix A . An $n \times n$ square matrix has exactly n eigenvalues (which may be real or complex, distinct or repeated). Eigenvalues are unique to a given matrix.

7.2.3 Eigenvectors

For each eigenvalue λ , there exists a corresponding non-zero column vector X that satisfies the homogeneous system of equations:

$$(A - \lambda I)X = 0$$

This vector X is called the eigenvector of A corresponding to the eigenvalue λ . Eigenvectors are not unique; if X is an eigenvector, then any scalar multiple kX (where $k \neq 0$) is also an eigenvector.

7.2.4 Working Rule for Eigenvalues and Eigenvectors

1. **Find Characteristic Equation:** Form

$$|A - \lambda I| = 0.$$

2. **Find Eigenvalues:** Solve the polynomial equation for the roots

$$\lambda_1, \lambda_2, \dots, \lambda_n.$$

3. **Find Eigenvectors:** For each root λ_i , substitute it back into the equation

$$(A - \lambda_i I)X = 0$$

. Solve this homogeneous system using row reductions to Echelon form to find the non-zero vector X .

7.2.5 Solved Example

Find the eigenvalues and eigenvectors for:

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

Step 1: Characteristic Equation

$$|A - \lambda I| = \begin{vmatrix} 5 - \lambda & 4 \\ 1 & 2 - \lambda \end{vmatrix} = 0$$

$$(5 - \lambda)(2 - \lambda) - 4 = 0 \implies \lambda^2 - 7\lambda + 6 = 0$$

Step 2: Eigenvalues Factoring the equation $(\lambda - 1)(\lambda - 6) = 0$ yields eigenvalues $\lambda = 1, 6$.

Step 3: Eigenvector for $\lambda = 1$

$$(A - 1I)X = \begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Row operations yield

$$x_1 + x_2 = 0$$

. Choosing

$$x_1 = 1$$

, we get

$$x_2 = -1.$$

Eigenvector

$$X_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

7.3 What Comes Next

8 Cayley Hamilton Theorem: State and Prove

Series: Matrix Theory **Lecture:** 09 **YouTube:** <https://www.youtube.com/watch?v=yJUoUn9f2Wo>

8.1 Prerequisites

- Knowledge of characteristic polynomials and characteristic equations.
- Familiarity with matrix cofactors and the adjoint matrix identity

$$A \cdot \text{adj}(A) = |A|I$$

8.2 Core Content

The Cayley-Hamilton theorem is a fundamental theorem in linear algebra that establishes an elegant relationship between a square matrix and its characteristic equation.

8.2.1 Statement

Every square matrix satisfies its own characteristic equation.

If A is a square matrix of order $n \times n$ and its characteristic equation is:

$$|A - \lambda I| = a_0 \lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \cdots + a_n = 0$$

Then replacing the scalar variable λ with the matrix A yields a valid matrix identity:

$$a_0 A^n + a_1 A^{n-1} + a_2 A^{n-2} + \cdots + a_n I = O$$

Where I is the identity matrix and O is the zero matrix of order n .

8.2.2 Proof Analysis

Let A be an $n \times n$ square matrix. Its characteristic equation is a polynomial of degree n in λ .

The elements of the characteristic matrix $(A - \lambda I)$ are polynomials in λ of degree at most 1. Consequently, the cofactors of $(A - \lambda I)$ are determinants of order $n - 1$, meaning they are polynomials in λ of degree at most $n - 1$.

Since the adjoint matrix $\text{adj}(A - \lambda I)$ is the transpose of the cofactor matrix, its elements are also polynomials of degree at most $n - 1$. We can express the adjoint matrix as a polynomial matrix in λ :

$$\text{adj}(A - \lambda I) = B_0 \lambda^{n-1} + B_1 \lambda^{n-2} + \cdots + B_{n-1}$$

Where $B_0, B_1, \dots, B_{n-1}$ are $n \times n$ matrices with scalar entries independent of λ .

Using the fundamental matrix property for any square matrix D ,

$$D \cdot \text{adj}(D) = |D| I$$

. Applying this to $D = A - \lambda I$:

$$(A - \lambda I) \cdot \text{adj}(A - \lambda I) = |A - \lambda I| I$$

Substitute the polynomial expressions:

$$(A - \lambda I)(B_0 \lambda^{n-1} + B_1 \lambda^{n-2} + \cdots + B_{n-1}) = (a_0 \lambda^n + a_1 \lambda^{n-1} + \cdots + a_n) I$$

By expanding the left-hand side and equating the matrix coefficients of corresponding powers of λ on both sides, we obtain a system of matrix equations. Pre-multiplying these equations by descending powers of A (i.e., $A^n, A^{n-1}, \dots, I$) and adding them together causes a telescoping cancellation of all terms involving B_i .

The remaining sum simplifies exactly to:

$$a_0 A^n + a_1 A^{n-1} + \cdots + a_n I = O$$

This completes the proof.

8.3 What Comes Next

9 Cayley Hamilton Theorem: Solved Problems

Series: Matrix Theory **Lecture:** 10 **YouTube:** <https://www.youtube.com/watch?v=FwpmB9cGSLE>

9.1 Prerequisites

- Understanding of the Cayley-Hamilton theorem statement.
- Proficiency in matrix multiplication and characteristic equation derivation.

9.2 Core Content

The Cayley-Hamilton theorem is a highly practical tool. Its most prominent application is computing the inverse of a large matrix (A^{-1}) efficiently by bypassing the need to compute the cofactor and adjoint matrices.

9.2.1 Finding the Inverse

According to the theorem, if the characteristic equation is:

$$a_0 A^n + a_1 A^{n-1} + \cdots + a_{n-1} A + a_n I = O$$

Assuming the matrix is non-singular (

$$|A| = a_n \neq 0$$

), we can multiply the entire equation by A^{-1} :

$$(a_0 A^n + a_1 A^{n-1} + \cdots + a_n I) A^{-1} = O$$

$$a_0 A^{n-1} + a_1 A^{n-2} + \cdots + a_{n-1} I + a_n A^{-1} = O$$

Rearranging the terms to isolate A^{-1} :

$$a_n A^{-1} = -(a_0 A^{n-1} + a_1 A^{n-2} + \cdots + a_{n-1} I)$$

$$A^{-1} = -\frac{1}{a_n} (a_0 A^{n-1} + a_1 A^{n-2} + \cdots + a_{n-1} I)$$

This allows us to compute A^{-1} using only basic matrix multiplications.

9.2.2 Solved Problem: Order 2 Matrix

Verify Cayley-Hamilton theorem and find A^{-1} for:

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

1. Characteristic Equation:

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ -1 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(3-\lambda) + 2 = 0 \implies \lambda^2 - 4\lambda + 5 = 0$$

2. Verification: We must show

$$A^2 - 4A + 5I = O.$$

$$A^2 = \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 8 \\ -4 & 7 \end{bmatrix}$$

$$A^2 - 4A + 5I = \begin{bmatrix} -1 & 8 \\ -4 & 7 \end{bmatrix} - \begin{bmatrix} 4 & 8 \\ -4 & 12 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O$$

3. Finding A^{-1} : Multiply the identity

$$A^2 - 4A + 5I = O$$

by A^{-1} :

$$A - 4I + 5A^{-1} = O \implies A^{-1} = \frac{1}{5}(4I - A)$$

$$4I - A = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{5} \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}$$

9.2.3 Solved Problem: Order 3 Matrix

Find A^{-1} using Cayley-Hamilton for:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

1. Characteristic Equation:

$$|A - \lambda I| = 0 \implies \lambda^3 - 6\lambda^2 + 7\lambda + 2 = 0$$

2. Cayley-Hamilton Application: The matrix satisfies

$$A^3 - 6A^2 + 7A + 2I = O.$$

Multiply by A^{-1} :

$$A^2 - 6A + 7I + 2A^{-1} = O \implies A^{-1} = \frac{1}{2}(-A^2 + 6A - 7I)$$

Compute the required matrix components:

$$A^2 = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}$$

$$-A^2 + 6A - 7I = -\begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} + \begin{bmatrix} 6 & 0 & 12 \\ 0 & 12 & 6 \\ 12 & 0 & 18 \end{bmatrix} - \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix} = \begin{bmatrix} -6 & 0 & 4 \\ -2 & 1 & 1 \\ 4 & 0 & -2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{2} \begin{bmatrix} -6 & 0 & 4 \\ -2 & 1 & 1 \\ 4 & 0 & -2 \end{bmatrix}$$

9.3 What Comes Next

This concludes the Matrix Theory lecture series. Mastering these concepts provides the foundational tools necessary for advanced applications in linear algebra.